

TESTING NORMALITY IN SMALL SAMPLES OF SELECTED SHAPE PARAMETER ESTIMATORS OF A GENERALIZED PARETO DISTRIBUTION

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Summary: To choose between estimators in small samples is often difficult. The added advantage of normality of an estimator implies that hypotheses can easily be tested. However, the bias of the estimator must also be taken into account in order to obtain correct results when hypotheses are tested assuming normality. In this paper, the normality, the bias and distribution of t -statistics formed of three popular estimators of the shape parameter of the generalized Pareto distribution were investigated in small samples. A confidence interval for the index of the S&P500 index is found by applying the results to estimators of the generalized Pareto distribution.

1. Introduction

Three well-known shape parameter estimators of a generalized Pareto distribution (GPD) are used to test hypotheses assuming normality in small samples. Acceptable sample sizes to assume normality are also investigated. An estimator can often be proven to have an asymptotically normal distribution. This can, however, be very vague and for some estimators this would mean large sample sizes of thousands of observations, which might not be available in practical problems. Also considered in this work is the case where the true parameter being estimated in a simulation is known, but not the distribution of the estimator in small samples. In this paper the question then investigated is whether correct conclusions will be made if a t -test is performed concerning the shape parameter.

The three estimators that are investigated are the probability weighted moment (PWM) estimator, the maximum likelihood (ML) estimator and the empirical Bayes estimator proposed by Zhang and Stephens (2009). Samples are simulated from the GPD. The GPD is also called a three parameter GPD with location parameter zero. The distribution of the Zhang and Stephens (2009) estimator is the closest to normally distributed in small samples and can be used with confidence to test hypotheses in smaller samples. The ML estimator converges fast to normality, but the t -statistic indicates that true hypotheses concerning the shape parameter will be rejected because of the large bias in the ML estimator in small samples. The PWM estimator is not normally distributed in small samples.

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An important application of the GPD is in risk analysis and finance and the success of its application depends on the quality of the estimators of the parameters (McNeil, 1997; Embrechts, Klüppelberg and Mikosch, 2003). In Section 4 it is shown that with 95% certainty, the index of the S&P500 is at least 2.7196, thus with a finite variance.

The Shapiro-Wilk test (Shapiro and Wilk, 1965) and the Anderson-Darling (AD) (Anderson and Darling, 1954) tests were used to test for normality. Normality as such does not imply good results with respect to bias and mean square error (MSE). The three estimators used in this study were shown to perform well (Zhang and Stephens, 2009; de Zea Bermudez and Kotz, 2010; Mackay, Challenor and Bahaj, 2011).

The t -test statistic will also be calculated. In a simulation study the true parameter is known and if the t -statistic is such that the hypothesis is rejected if normality was confirmed, it will be an indication that the estimator is biased. If no assumption is made about the distribution and the null-hypothesis is rejected it can be an indication of non-normality, bias or both. This is an exact method although very simple, and under the hypothesis of normality will involve no approximations when tests are performed.

Consider a general estimation problem of a parameter θ in a simulation study. Suppose m samples of size n each are generated from a distribution with true known parameter θ_0 , and estimators, $\hat{\theta}_1, \dots, \hat{\theta}_m$ of θ_0 , calculated. To test the normality of the estimator, the following statistic can be used:

$$t_{m-1} = \frac{(m-1)(\bar{\theta} - \theta_0)}{\sqrt{S_{\bar{\theta}}^2}}, \quad \bar{\theta} = \frac{1}{m} \sum_{j=1}^m \hat{\theta}_j, \quad S_{\bar{\theta}}^2 = \frac{1}{m} \sum_{j=1}^m (\hat{\theta}_j - \bar{\theta})^2. \quad (1)$$

If this statistic is used to test the hypothesis, $H_0 : \theta = \theta_0$, it will be rejected with high probability only if the estimator is not normally distributed or if there is a large bias, because the hypothesis is true and the estimated variance correct.

The estimated bias and MSE using the true parameter are: $B(\hat{\theta}) = \bar{\theta} - \theta_0$, and $MSE = \frac{1}{m} \sum_{j=1}^m (\hat{\theta}_j - \theta_0)^2$. It follows that $mMSE = vS_{\bar{\theta}}^2 + mB^2(\hat{\theta})^2$, $v = m - 1$.

$S_{\bar{\theta}}^2 = \frac{1}{m} \sum_{j=1}^m (\hat{\theta}_j - \theta_0)^2$ and the statistic in (1) can be written in terms of the bias and MSE as

$$t = \frac{\bar{\theta} - \theta_0}{S(\bar{\theta})} = \sqrt{v}B(\hat{\theta}) / \sqrt{(MSE - B^2(\hat{\theta}))}, \quad v = m - 1. \quad (2)$$

The number of simulated samples m will mostly be large, more than say 500 samples generated, thus the t -statistic is approximately standard normally distributed with $t \sim N(0, 1)$.

Another way to use the t -statistic form in a simulation study is where an expression for the standard value of the estimator is available and estimators are calculated from say m generated samples and then to use a goodness of fit test using the calculated t values. Thus testing if the calculated statistics, $t_j = (\hat{\theta}_j - \theta_0)/S_{\hat{\theta}}$, $j = 1, \dots, m$, fit the standard t -distribution. For example if the Kolmogorov-Smirnov test is used, since the distribution of t_j , $j = 1, \dots, m$, involves no unknown parameters and the exact test statistic can be calculated.

Using the statistic written in terms of the estimated bias and MSE at sample size n , means that previously published results where the MSE and bias are reported can be used to check the normality of an estimator. An application of using the t -statistic to test normality of GPD estimator parameters using previously published results (where the MSE and bias are reported), is given in Section 3.3.

2. The GPD and estimation of its parameters

The three-parameter GPD (Johnson, Kotz and Balakrishnan, 1994) distribution is

$$F(x) = 1 - (1 + (\xi/\sigma)(x - \mu))^{-1/\xi}, \quad x - \mu \geq 0, \sigma > 0,$$

σ a scale parameter, ξ the shape parameter and μ the location parameter. For $\mu = 0$ it reduces to a two-parameter distribution and will be referred to as a GPD distribution. In peaks over threshold problems, a threshold $\hat{\mu}$ is estimated, and estimation performed on the excesses over the threshold which are assumed to be GPD distributed. The index is defined as $\alpha = 1/\xi$, and determines the tail heaviness. The distribution reduces to the exponential distribution for $\xi = 0$. The domain of the distribution is finite for $\xi < 0$, and in that case, the domain is $(\mu, \mu - \sigma/\xi)$.

The approximate normality of the PWM, Zhang and Stephens (2009) and two-parameter ML estimators of the shape parameter ξ in smaller samples is investigated. Using the PWM procedure means that the parameters are estimated as

$$\begin{aligned} \hat{\xi} &= 2 - \hat{a}_0/(\hat{a}_0 - 2\hat{a}_1) \quad \text{and} \quad \hat{\sigma} = 2\hat{a}_0\hat{a}_1/(\hat{a}_0 - 2\hat{a}_1), \\ \hat{a}_0 &= \bar{x}, \quad \hat{a}_1 = \frac{1}{n} \sum_{j=1}^n x_{(j)}(1 - p_j), \quad p_j = (j - 0.35)/n, \end{aligned}$$

for a sample of size n from the two-parameter GPD and $x_{(1)} \leq \dots \leq x_{(n)}$ the order statistics (de Zea Bermudez and Kotz, 2010).

Three other possible applications of the t -statistic are given below. In order to use the t -statistic to investigate properties of an estimator $\hat{\theta}$ it must first be established that the estimator is approximately normally distributed at the sample size of interest, say n . The bias of an estimator at sample size n should be taken into account when using a t -statistic and this aspect should also be checked.

- In practical problems it is sometimes important to know at which sample size an approximation of the variance or an asymptotic variance, $\hat{\sigma}_n^2$, of an estimator is valid to use in inferential aspects concerning the parameter $\theta = \theta_0$. The m t -statistics formed in a simulation study to test the true null hypothesis, $H_0 : \theta = \theta_0$, $t_j = (\hat{\theta}_j - \theta_0)/\hat{\sigma}_n$, $j = 1, \dots, m$, should be approximately normally distributed if the variance is valid, with mean 0 and variance 1. A goodness of fit test can be used to check if this is true, and if so, the variance $\hat{\sigma}_n^2$ is valid to use at that sample size. An example is the estimated variance S^2 of normal data. Even if S^2 is approximately normally distributed for a sample sizes of say $n = 250$, the approximate estimate of its variance $\text{var}(S^2) = 2S^4/(n - 1)$, may not yet be valid to use in hypothesis testing at this sample size.
- If only a single sample of size n is available, and not m samples as with a simulation study, a bootstrap estimate of the variance of the estimator $\hat{\theta}$ of $\theta = \theta_0$, can be calculated, $\hat{\sigma}_b^2$ using m bootstrap samples. The m t -statistics, $t_j = (\hat{\theta}_j - \theta_0)/\hat{\sigma}_b$, $j = 1, \dots, m$, are formed to test the hypothesis: $H_0 : \theta = \theta_0$. Each of the statistics, t_j , $j = 1, \dots, m$, should be approximately normally distributed under the null hypothesis, with mean 0 and variance 1. The statistic, $t_{m-1}^* = (m - 1)\bar{t}/\sqrt{\hat{\sigma}_b^2}$, with $\bar{t} = \frac{1}{m} \sum_{j=1}^m t_j$, $\hat{\sigma}_b^2 = \frac{1}{m} \sum_{j=1}^m (t_j - \bar{t})^2$, should be t distributed with $m - 1$ degrees of freedom. Alternatively, a goodness of fit test can be applied to the bootstrap estimated sample, $\hat{\theta}_1, \dots, \hat{\theta}_m$.

- In Section 3.3 it is shown that writing a t -statistic in terms of MSE and bias can be used to test normality by making use of previously published simulation results, where the MSE and bias of an estimator were reported. Using (2) a large t -statistic would indicate that the estimator is not normally distributed or with a large bias with respect to the true parameter, given the MSE and bias.

The simulation study has two parts. Firstly, a simulation testing normality for various samples sizes and shape parameters, and secondly the simulation results given in the paper of Hosking and Wallace (1987) are used to show how previously published results can be used to check normality of three estimators in small samples. In Section 3.1 three methods to test normality are used to test the normality of the PWM, Zhang and Stephens (2009) and ML estimators of the shape parameter of the GPD with respect to sample size.

3. Simulation study

3.1. Normality of the GPD shape parameter estimators

In this section a simulation study is reported to investigate the approximate normality of the PWM, Zhang and Stephens (2009) and two-parameter GPD ML shape parameter (ξ) estimators. Data are simulated from the GPD with $\sigma = 1$, $\mu = 0$, $\xi = -0.25, 0.1, 0.25, 0.5, 0.75$, and sample sizes $n = 50, 100, 250, 500$. The results are based on $m = 1000$ samples.

In Table 1 the mean square error (MSE) and bias is shown for the three estimators for various sample sizes and value of the shape parameter. Bias is calculated as $\hat{\xi} - \xi$.

The results when testing the distribution of the estimators for normality is shown in Table 2. Especially in smaller samples the Zhang and Stephens (2009) estimator outperforms the other estimators with respect to bias and MSE. The simulation study was repeated to investigate the normality of the estimators.

The results in Table 2 show that the estimator of Zhang and Stephens (2009) is approximately normally distributed for $n \geq 50$, $\xi > 0$ and for $\xi < 0$, the sample size must be larger than 250 for approximate normality. The PWM estimator is normally distributed for $\xi = 0.1, 0.25$ and $n \geq 250$. The ML estimator converges fast to normality with respect to sample size for positive values of ξ and the hypothesis of normality can be accepted at a sample size of $n = 100$ for small values of ξ and for $\xi \geq 0.25$ at a sample size of $n = 50$. For negative values of ξ the ML estimator is normally distributed for $n > 250$.

It can be seen from the t -statistic results that the ML estimator would reject the hypothesis, $H_0 : \xi = \xi_0$, when a t -test is used and the data simulated with parameter $\xi = \xi_0$. This is probably because the ML estimator is very biased in smaller samples even if the estimator is normally distributed. A sample size of say $n > 500$ will be needed before the t -test can be used with confidence to test hypothesis concerning ξ . The Zhang and Stephens (2009) estimator is not only normally distributed, but gives the correct results when the t -test is applied.

Table 1: MSE and bias for various sample sizes and values of the shape parameter.

	PWM		Zhang-Stephens		ML	
	Bias	MSE	Bias	MSE	Bias	MSE
$\xi = -0.25, \sigma = 1.0$						
50	-0.0244	0.0347	0.024	0.0215	-0.075	0.0301
100	-0.0094	0.0176	0.0138	0.0092	-0.0392	0.0113
250	-0.0028	0.0065	0.0069	0.0029	-0.0159	0.0032
500	-0.0027	0.0032	0.0033	0.0013	-0.0089	0.0015
$\xi = 0.1, \sigma = 1.0$						
50	-0.0286	0.0286	0.0215	0.0286	-0.0496	0.0354
100	-0.0155	0.0137	0.012	0.0131	-0.0251	0.0148
250	-0.0063	0.0055	0.0051	0.0052	-0.0103	0.0055
500	-0.0025	0.0026	0.0036	0.0024	-0.0041	0.0025
$\xi = 0.25, \sigma = 1.0$						
50	-0.0358	0.0272	0.0217	0.0307	-0.036	0.0364
100	-0.023	0.0156	0.0074	0.0166	-0.024	0.0186
250	-0.0103	0.0059	0.0026	0.0061	-0.0098	0.0065
500	-0.003	0.0033	0.0035	0.0034	-0.0026	0.0035
$\xi = 0.5, \sigma = 1.0$						
50	-0.0887	0.0395	-0.0039	0.0443	-0.0431	0.0534
100	-0.0557	0.0187	-0.0031	0.0189	-0.0242	0.021
250	-0.0259	0.0101	-0.0012	0.0085	-0.0091	0.0089
500	-0.0204	0.0061	-0.0026	0.0046	-0.0065	0.0047
$\xi = 0.75, \sigma = 1.0$						
50	-0.1718	0.0591	-0.0243	0.0565	-0.0474	0.0667
100	-0.1177	0.0322	-0.0078	0.0284	-0.0191	0.031
250	-0.0881	0.0175	-0.0079	0.0118	-0.0119	0.0123
500	-0.0629	0.0118	-0.0037	0.0063	-0.0057	0.0064

3.2. An application to excesses over a threshold

In this section samples were simulated from the three-parameter GPD with $\mu = 1$, $\sigma = 1$. The threshold was estimated as the smallest observation in the sample and estimation was performed on the excesses over that threshold. In Figure 1 a histogram of estimated parameters $\hat{\xi}$ of $\xi = 0.75$, using the Zhang and Stephens (2009) method, is shown.

The skewness and kurtosis of the three estimators are respectively, 0.2238, 3.4435 for PWM, 0.0167, 3.1591 for Zhang and Stephens, and 0.0124, 3.4413 for ML.

In the next simulation study the testing of hypotheses for excesses over a threshold, using the Zhang and Stephens (2009) estimator and a bootstrap estimate for the variance is investigated. There are $m = 1000$ simulated statistics and each time the statistic $z = (\hat{\xi} - \xi_0) / \hat{\sigma}_*(\hat{\xi})$ is calculated, where $\hat{\sigma}_*^2(\hat{\xi})$ denotes the bootstrap variance for each sample. Estimation is performed on excesses over a threshold and the original samples simulated from a three-parameter GPD with $\mu = 1$. The number

Table 2: Testing normality of three shape parameter estimators of a GPD, for various sample sizes and parameters. P -values given and normality is rejected for $p < 0.05$. Based on $m = 1000$ samples generated for each sample size and set of parameters.

	PWM			Zhang-Stephens			ML		
	Shapiro-Wilk p -value	AD p -value	p -value and t -statistic	Shapiro-Wilk p -value	AD p -value	p -value and t -statistic	Shapiro-Wilk p -value	AD p -value	p -value and t -statistic
$\xi = -0.25, \sigma = 1.0$									
50	0.0000	0.0000	0.0000 (-4.1026)	0.0172	0.0184	0.0000 (5.0716)	0.0001	0.0003	0.0000 (-15.2522)
100	0.0000	0.0000	0.003 (2.9768)	0.0004	0.0002	0.0000 (4.3245)	0.0002	0.0000	0.0000 (-12.5349)
250	0.0000	0.0000	0.1507 (-1.4381)	0.0059	0.1453	0.0024 (3.0483)	0.0071	0.1339	0.0000 (-10.5779)
500	0.0503	0.1717	0.9609 (0.0490)	0.7463	0.3159	0.0000 (4.1762)	0.6075	0.4103	0.0000 (-6.1586)
$\xi = 0.1, \sigma = 1.0$									
50	0.0000	0.0000	0.0000 (-7.1942)	0.0264	0.2720	0.1093 (1.6028)	0.0009	0.0470	0.0000 (-11.2279)
100	0.0223	0.0537	0.0000 (-6.3750)	0.9858	0.3201	0.1811 (1.3382)	0.6511	0.1264	0.0000 (-8.9987)
250	0.4463	0.1943	0.0005 (-3.4735)	0.4703	0.2331	0.1627 (1.3969)	0.4712	0.2472	0.0000 (-5.2906)
500	0.1965	0.0130	0.0023 (-3.0554)	0.6944	0.6063	0.6962 (0.3395)	0.6401	0.6835	0.0000 (-4.5390)
$\xi = 0.25, \sigma = 1.0$									
50	0.0022	0.0008	0.0000 (-7.7543)	0.5519	0.5068	0.0129 (2.4917)	0.7115	0.4851	0.0000 (-7.7023)
100	0.0017	0.0041	0.0000 (-6.6671)	0.6120	0.8875	0.1461 (1.4544)	0.6340	0.8063	0.0000 (-5.9528)
250	0.1454	0.8577	0.0003 (-3.5890)	0.8538	0.5500	0.1687 (1.3775)	0.6443	0.4780	0.0009 (-3.3258)
500	0.5646	0.2264	0.0014 (-3.2051)	0.9694	0.5991	0.8337 (0.2101)	0.9343	0.6450	0.0016 (-3.1568)
$\xi = 0.5, \sigma = 1.0$									
50	0.4047	0.9032	0.0000 (-15.4597)	0.8210	0.4964	0.7246 (-0.3524)	0.4949	0.3853	0.0000 (-5.8278)
100	0.0261	0.0589	0.0000 (-14.3368)	0.7905	0.4177	0.0136 (-2.4732)	0.8812	0.4429	0.0000 (-6.8130)
250	0.0000	0.0000	0.0000 (-8.0037)	0.6387	0.2691	0.4536 -0.7497	0.6488	0.2747	0.0635 (-1.8578)
500	0.0000	0.0000	0.0000 (-9.2181)	0.8647	0.1396	0.2082 (-1.2594)	0.8835	0.1417	0.002 (-3.1008)
$\xi = 0.75, \sigma = 1.0$									
50	0.0002	0.0019	0.0000 (-31.2968)	0.3077	0.1603	0.0004 (-3.5619)	0.7866	0.3813	0.0000 (-6.0058)
100	0.0480	0.4749	0.0000 (-30.2926)	0.2885	0.432	0.0085 (-2.6381)	0.4247	0.4861	0.0000 (-4.6252)
250	0.0000	0.0000	0.0000 (-23.4367)	0.2617	0.2457	0.2490 (-1.1536)	0.3564	0.3050	0.0264 (-2.2242)
500	0.0000	0.0000	0.0000 (-23.6112)	0.6104	0.6001	0.2850 (-1.0677)	0.5963	0.5984	0.0661 (-1.8395)

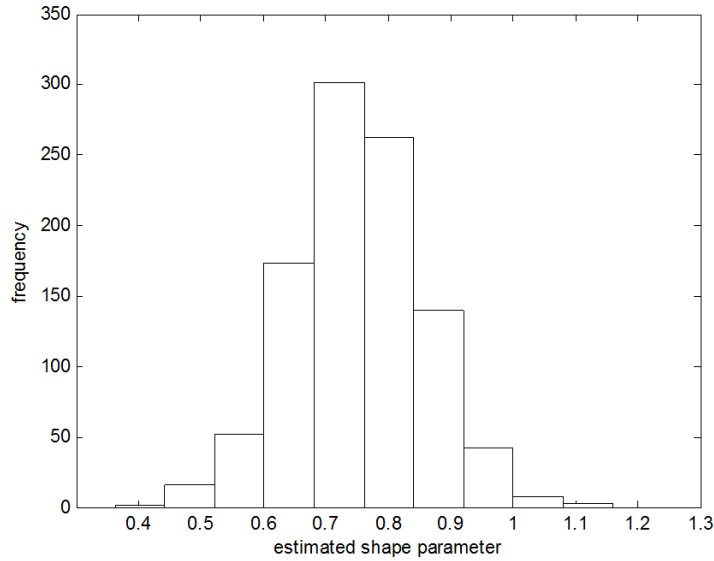


Figure 1: Histogram of $m = 1000$, $n = 250$ estimated shape parameter from GPD with $\xi = 0.75$, $\sigma = 1.0$, $\mu = 1.0$. Zhang and Stephens (2009) estimator.

of rejections at the 5% level is shown in Table 3.

With respect to the type I error, one can use the assumption of normality and test hypotheses concerning the shape parameter, for the range of ξ and sample sizes considered.

3.3. Using previously published results to check normality

Hosking and Wallace (1987) conducted a simulation study, generating $m = 50000$ samples to estimate the MSE and bias of the maximum likelihood estimator (MLE), method of moments (MOM) and PWM estimators (Tables 2 and 3 in their paper). They reported the estimated bias and RMSE, where MSE is the square of RMSE. Using the equation

$$z \approx \sqrt{v}B(\hat{\theta}) / \sqrt{(MSE - B^2(\hat{\theta}))} = \sqrt{v}B(\hat{\theta}) / \sqrt{(RMSE^2 - B^2(\hat{\theta}))},$$

where $v = m - 1$, their results are used to check for normality in small samples in this paper (Table 4).

The estimators are not normally distributed for samples sizes less than $n = 100$. The fast convergence of the ML estimator to normality can also be seen by noting the rate of decrease of z with respect to sample size.

Table 3: Checking the rejection rate under the true hypothesis at the 5% level using the method of Zhang and Stephens (2009) and a bootstrap estimate of the variance.

Testing: $H_0 : \xi = \xi_0$		
n	Mean, variance z	% rejected
$\xi_0 = 0.2, \sigma = 1.0$		
15	0.0475 (0.9196)	5.1
50	0.0411 (1.1021)	6.4
100	0.0189 (1.2174)	5.7
150	0.0229 (1.1381)	6.8
250	0.0204 (1.1145)	5.9
$\xi_0 = 0.4, \sigma = 1.0$		
15	0.0187 (0.9134)	4.9
50	0.0859 (1.1436)	6.7
100	0.0579 (1.1306)	5.7
150	0.0883 (1.1467)	7.7
250	0.0820 (1.1364)	5.8
$\xi_0 = 0.6, \sigma = 1.0$		
15	0.2166 (1.1077)	7.5
50	0.1601 (1.2994)	8.1
100	0.1974 (1.2195)	7.8
150	0.1195 (1.1481)	6.9
250	0.1698 (1.1313)	7.1

Table 4: Test statistics for normality calculated using reported bias and MSE of Hosking and Wallace (1987).

$\sigma = 1, \xi = 0.4$									
n	ML			MOM			PWM		
	Bias	RMSE	z	Bias	RMSE	z	Bias	RMSE	z
15	0.16	0.46	82.9561	0.30	0.38	287.6118	0.18	0.36	129.0995
50	0.05	0.22	52.1854	0.17	0.21	308.3274	0.07	0.19	88.6147
100	0.02	0.15	30.0828	0.13	0.13	311.6512	0.04	0.14	66.6667

4. A confidence interval for the log returns of the S&P500

Using the closing values of the S&P500 over 5 years (2006 to 2011), 1 258 log returns are calculated. A GPD is fitted to the largest 150 returns using a threshold of 0.0060. The purpose is to calculate a 95% confidence interval for ξ using the above approximate normality of the Zhang and Stephens (2009) estimator and a bootstrap estimate for the variance of the estimator. The estimated values of σ using MOM, PWM and Zhang and Stephens (2009) estimators are 0.0046, 0.0044, and 0.0044 respectively, and the estimated values of ξ using the three methods are respectively 0.1494, 0.1752, and 0.1919.

The bootstrap estimate of the standard deviation of the Zhang and Stephens (2009) estimate of ξ is 0.0897, resulting in a 95% confidence interval [0.0162: 0.3677], with the implication that the index of the tail observations is not less than 2.7196(= 1/0.3677). The P-P plot of the estimated distribution versus the empirical distribution function is shown in Figure 2.

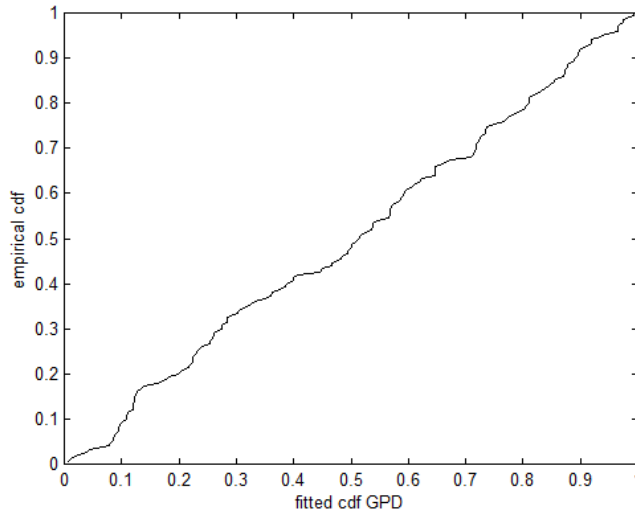


Figure 2: P-P plot of the fitted GPD and empirical distribution to the 150 largest log returns from a sample of size 1258. The line is close to straight and the GPD gives a good fit.

5. Conclusions and remarks

Normality in smaller samples of the shape parameter estimators of a GPD is investigated. The estimator of Zhang and Stephens (2009) is shown to be close to normally distributed even in small samples when fitting the GPD to excesses over a threshold. The knowledge of a normally distributed estimator, can assure more confidence in conclusions when testing hypotheses. The ML estimator is normally distributed even in sample of say $n = 100$ for positive ξ , but have a large bias in smaller

samples, which causes the testing of the hypotheses concerning the true parameter to be incorrectly rejected.

The MOM, PWM and ML can be considered as the building blocks of the other estimation methods (Mackay et al., 2011). There are overview papers on the performance of estimators of the GPD and a wide variety of estimators are considered in the papers of de Zea Bermudez and Kotz (2010) and Mackay et al. (2011).

Hosking and Wallace (1987) focused more on the comparison of ML and PWM estimation, and according to their work the PWM method compares favourably to ML in problems where ξ is small, while ML outperforms the PWM method for samples larger than 500. They also found that the ML estimator does not achieve the Cramer-Rao lower bound for sample sizes up to 500. In the present paper the ML estimator is approximately normally distributed for a sample size of 100, but very biased for samples up to approximately $n = 500$. Convergence problems sometimes occur when using ML estimation in small samples and especially when ξ is close to zero or negative.

Mackay et al. (2011) found that the Zhang and Stephens (2009) estimator was not very sensitive to the choice of the threshold in peaks over threshold problems. It compared favourably to the ML estimator also in larger sample sizes, always existed, and was much easier to work with from a computational viewpoint than ML estimation. Taking these factors into account and the fact that it can be used to test hypotheses, it would seem that the Zhang and Stephens (2009) estimator can be recommended as the most stable estimator of the index. If the sample size is larger than 500 and if computational capacity is available, it would be good to confirm the results using the ML estimator.

The use of a t -statistic in a simulation study where the parameter tested is known and the hypothesis is true, can give an indication of non-normality, but bias of the estimator especially in small samples will also influence the statistic. The t -statistic is formed using the true parameter in a simulation study using a correct estimated standard deviation, and would make this statistic a good indicator of how well hypotheses testing assuming normality will perform in practical problems.

The t -statistic can be applied to previously published results of a simulation study where the estimated MSE and bias were given. If the t -statistic formed using the true parameter and equation (2), is too large and lead to rejection of the true hypothesis it would be an indication of non-normality and/or bias in the estimator, and will show if hypotheses assuming normality can be tested at the specific sample sizes considered.

An estimator can be asymptotically normally distributed and an expression for the asymptotic variance could be known (Hosking and Wallace, 1987). The consideration of at which sample sizes asymptotic variances are accurate enough to be of practical value for use in a t -test is not considered in this work. This is an important question and needs further investigation.

The rate of decrease of the t -statistic with respect to sample size can also give an indication of the rate of convergence to normality of the estimator. In other words if the absolute value of the t -statistic is modelled as a function of n , of say $\alpha n^{-\beta}$, the estimated β would give an indication of the rate of convergence to normality of the specific estimator in terms of the sample size.

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